

Domination and dominator colorings in planar graphs with small diameter

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Abstract

A dominator coloring of a graph G is a proper coloring of the vertices of G in which each vertex of the graph dominates every vertex of some color class, where a vertex dominates itself and all vertices adjacent to it. The dominator chromatic number of G is the minimum number of colors among all dominator coloring of G . A total dominator coloring of a graph G is a proper coloring of the vertices of G in which each vertex of the graph dominates every vertex of some color class other than its own. The total dominator chromatic number of G is the minimum number of colors among all total dominator coloring of G . In this paper, we present bounds on the dominator chromatic number and total dominator chromatic number of a planar graphs with small diameter. In particular we show the dominator chromatic number of a planar graph of diameter 2 is at most 5. We also present results for the special cases of outerplanar graphs and bipartite planar graphs.

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1 Introduction

A vertex in a graph G *dominates* itself and all vertices adjacent to it. A *dominating set* of G is a set $S \subseteq V(G)$ such that every vertex in $V(G) \setminus S$ is adjacent to at least one vertex in S , that is, every vertex of G is dominated by some vertex of the dominating set. The *domination number* of G , denoted by $\gamma(G)$, is the minimum cardinality of a dominating set of G . A dominating set of G of cardinality $\gamma(G)$ is called a γ -*set* of G . The concept of domination in graphs is now well studied. For fundamentals of domination theory in graphs we refer the reader to the recent book [9].

A *total dominating set*, abbreviated a TD-set, of a graph G with no isolated vertex is a set $S \subseteq V(G)$ such that every vertex in $V(G)$ is adjacent to at least one vertex in S . The *total domination number* of G , denoted by $\gamma_t(G)$, is the minimum cardinality of a TD-set of G . A TD-set of G of cardinality $\gamma_t(G)$ is called a γ_t -set of G . The literature on the subject of total domination in graphs has been surveyed and detailed in the book [16].

A *proper vertex coloring* of a graph G is an assignment of colors (elements of some set) to the vertices of G , one color to each vertex, so that adjacent vertices are assigned distinct colors. A proper vertex coloring whose colors are taken from a set of k colors, usually the set $[k] = \{1, 2, \dots, k\}$, is called a *proper k -coloring*. In a given coloring of G , a *color class* of the coloring is a set consisting of all those vertices assigned the same color. The *vertex chromatic number* of G , denoted $\chi(G)$, is the smallest positive integer k for which G has a proper k -coloring. A χ -*coloring* of G is a proper k -coloring of G that uses $\chi(G)$ colors. In what follows, we simply call a proper vertex coloring a *coloring* and we refer to the vertex chromatic number as the *chromatic number*.

Combining the concept of domination (total domination) in graphs with the concept of colorings in graphs, we get the concept of dominator colorings (respectively, total dominator colorings) of a graph, which we define formally as follows.

A vertex is a *dominator* of a set S in a graph G if it dominates every vertex in S . A *dominator coloring* of a graph G is a proper coloring of G with the additional property that every vertex in $V(G)$ dominates all vertices in at least one color class, that is, each vertex of the graph belongs to a singleton color class or is adjacent to every vertex of some (other) color class. The *dominator chromatic number* $\chi_d(G)$ of G is the minimum number of color classes in a dominator coloring of G . A χ_d -*coloring* of G is a dominator coloring of G that uses $\chi_d(G)$ colors.

The total version of dominator coloring in a graph is defined as follows. A *total dominator coloring*, abbreviated TD-*coloring*, of a graph G with no isolated vertex is a proper coloring of G in which each vertex of the graph is adjacent to every vertex of some other color class (different from its own color class). The *total dominator chromatic number* of G , which we denote by $\chi_{td}(G)$ (and denoted by $\chi_d^t(G)$ in [14, 17]) is the minimum integer k for which G has a TD-coloring with k colors. A χ_{td} -*coloring* of G is a dominator coloring of G that uses $\chi_{td}(G)$ colors.

The concept of a dominator coloring in a graph was birthed in the late 1970s when Cockayne, Hedetniemi and Hedetniemi [3] defined the domatic number of a graph involving partitions into dominating sets. The concept of total dominator colorings in graphs was first defined in the manuscript by Hedetniemi, Hedetniemi, McRae, Rall, and Hedetniemi [12] dated July 9, 2009. Dominator and total dominator colorings have been studied, for example, in [1, 2, 6, 7, 13, 15, 19].

2 Main results

In this paper we study dominator and total dominating colorings in planar graphs of small diameter, where the *diameter* of a graph G , denoted by $\text{diam}(G)$, is the maximum distance among all pairs of vertices of G . Along the way we provide results on the domination and total domination numbers of such graphs, especially for the case restricted to bipartite or outerplanar graphs.

As observed in [8], a tree of diameter 4 can have arbitrarily large domination number, implying (see Theorem 8) that its (total) dominator chromatic number can be arbitrarily large. The interesting question is what happens when the diameter of a planar graph is 2 or 3. As pointed out by MacGillivray and Seyffarth [18], the restriction of bounding the diameter on the domination number of a planar graph is reasonable to impose because planar graphs with small diameter are often important in applications such as interconnection networks (see [5]). We shall prove the following results.

Theorem 1 *If G is an outerplanar graph, then the following hold.*

- (a) *If $\text{diam}(G) \leq 2$, then $\gamma_t(G) \leq 3$ and $\chi_{td}(G) \leq 4$, and these bounds are best possible.*
- (b) *If $\text{diam}(G) \leq 3$, then $\gamma_t(G) \leq 4$, $\chi_d(G) \leq 5$, and $\chi_{td}(G) \leq 6$, and these bounds are best possible.*

Theorem 2 *If G is a bipartite planar graph with $\text{diam}(G) \leq 3$, then $\gamma_t(G) \leq 4$ and $\chi_{td}(G) \leq 6$.*

Theorem 3 *Apart from the graph F shown in Figure 1, a planar graph G of diameter 2 has a dominating pair $\{u, v\}$ such that $G - \{u, v\}$ is 3-colorable.*

Theorem 4 *If G is a planar graph of diameter 2, then $\chi_d(G) \leq 5$, and this bound is best possible.*

Theorem 1 is proven in Section 4 and Theorem 2 in Section 5. In Section 6 we use Theorem 3 to prove Theorem 4, and in Section 7 we consider planar graphs of diameter 3. Finally we provide the lengthy proof of Theorem 3 in Section 8. We start, however, with a discussion of previous results.

3 Known results

MacGillivray and Seyffarth [18] initiated the study of domination in graphs of bounded diameter. As they observed, the domination number of graphs of diameter two can be arbitrarily large. In contrast, they established that the domination number of planar graph of diameter two is at most 3. This bound was subsequently improved in [8].

Theorem 5 ([8]) *If G is a planar graph of diameter 2, then $\gamma(G) \leq 2$, except for the graph F of Figure 1 with domination number 3, and $\gamma_t(G) \leq 3$.*

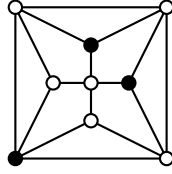


Figure 1: The planar graph F of diameter 2 with domination number 3

MacGillivray and Seyffarth [18] also showed that:

Theorem 6 ([18]) *If G is an outerplanar graph, then the following hold.*

- (a) *If G has diameter at most 2, then $\gamma(G) \leq 2$.*
- (b) *If G has diameter at most 3, then $\gamma(G) \leq 3$.*

MacGillivray and Seyffarth [18] proved that planar graphs of diameter 3 have domination number at most 10. Dorfling et al. [4] improved this result as follows.

Theorem 7 ([4]) *If G is a planar graph of diameter 3, then the following hold.*

- (a) *We have $\gamma(G) \leq 9$. Moreover, if G has sufficiently large order then $\gamma(G) \leq 6$.*
- (b) *We have $\gamma_t(G) \leq 10$. Moreover, if G has sufficiently large order then $\gamma_t(G) \leq 7$.*

We consider next results on the (total) dominator chromatic number. Since every vertex is a dominator of itself, the coloring of graph G that assigns a unique color to each vertex is a trivial dominator coloring of G . Thus, every graph G has a dominator coloring, and therefore the dominator chromatic number $\chi_d(G)$ is well-defined. Every total dominator coloring is a dominator coloring, and so $\chi_d(G) \leq \chi_{td}(G)$. Further, the (total) dominator chromatic number is intimately connected with the chromatic number and the (total) domination number. In particular we have the following fundamental bounds.

Theorem 8 *If G is a graph, then the following hold.*

- (a) $([6, 7]) \max\{\gamma(G), \chi(G)\} \leq \chi_d(G) \leq \gamma(G) + \chi(G).$
- (b) $([17, 19]) \max\{\gamma_t(G), \chi(G)\} \leq \chi_{td}(G) \leq \gamma_t(G) + \chi(G).$

More generally, if J is a (total) dominating set of graph G , then the (total) dominator chromatic number of G is at most $|J| + \chi(G - J)$, since one can give every vertex of J a unique color and give the remaining vertices any proper coloring.

4 Outerplanar graphs of bounded diameter

Recall that an outerplanar graph is a planar graph that can be drawn with all vertices on the outer face. It is well known that every outerplanar graph is 3-colorable. Theorem 1 follows from the propositions and theorem proven below.

4.1 Domination in outerplanar graphs

It follows from Part 1 of Theorem 6 that the total domination number of an outerplanar graph of diameter 2 is at most 3 (by adding the common neighbor of the dominating pair if needed). An example graph achieving this value is C_5 . However, this is the only example, as we now show.

Proposition 9 *If G is an outerplanar graph of diameter 2, the $\gamma_t(G) \leq 2$ except for C_5 .*

Proof. Consider an embedding of graph G in the plane such that all vertices lie on the unit circle $\mathcal{C}$ and all edges lie within $\mathcal{C}$. If there is a chord, then the ends of the chord form a TD-set of G , since vertices on either side of the chord have to reach each other. If there is no chord, then the graph G is a path or cycle, and the result can be checked. $\square$

Theorem 10 *If G is an outerplanar graph of diameter 3, then $\gamma_t(G) \leq 4$. Furthermore, the largest such graph achieving this value has order 10.*

Proof. Consider an embedding of G in the plane such that all vertices lie on the unit circle $\mathcal{C}$ and all edges lie within $\mathcal{C}$. If no edge is a chord of $\mathcal{C}$, then graph G is a cycle or a path. A path of diameter 3 has total domination number 2. A cycle of diameter 3 has at most 7 vertices and therefore has total domination number at most 4 (achieved for both the 6- and 7-cycle).

So assume there is an edge st of G that represents a chord of $\mathcal{C}$. This edge creates two subgraphs G_1 and G_2 of G such that both contain s and t , every other vertex is in exactly one of the subgraphs, and there is no edge between $V(G_1 - \{s, t\})$ and $V(G_2 - \{s, t\})$.

By the diameter condition, it cannot be that both subgraphs G_1 and G_2 contain a vertex at distance two from $\{s, t\}$. So assume the pair $\{s, t\}$ dominates G_1 . We are done unless there is vertex x in G_2 not dominated by $\{s, t\}$. There are two cases.

Case 1. Assume that G_1 contains both a vertex s' not adjacent to t and a vertex t' not adjacent to s . Necessarily, s' is adjacent to s and t' is adjacent to t . It follows that x is distance 2 from both s and t . Indeed every vertex in G_2 must be within distance 2 of both s and t .

First subcase: Assume vertices s and t have a common neighbor c inside G_2 . We claim that the set $\{s, t, c\}$ dominates G . We know that $\{s, t\}$ dominates all of G_1 . Let z be a vertex of G_2 not in the set. Without loss of generality we may assume it lies between s and c on $\mathcal{C}$. In order for it to reach t in two steps, vertex z must be adjacent to one of s and c . The claim follows.

Second subcase: Assume vertices s and t do not have a common neighbor inside G_2 . So assume x reaches s via vertex a and reaches y via vertex b . Any vertex that lies on $\mathcal{C}$ between s and a must be adjacent to s in order to reach t in 2 steps. There cannot be a vertex between a and x , nor between x and b . And any vertex between b and t must similarly be adjacent to t . Thus the pair $\{s, t\}$ dominates all of G except for x . It follows that for example $\{s, t, a\}$ is a total dominating set of G .

Case 2. Assume there is no chord for which Case 1 holds. It follows that for every chord st , at least one of the two ends dominates all of G_1 . Given such a chord st , we define its *zone* as the arc of $\mathcal{C}$ with endpoints s and t that contains all of G_1 . Since chords do not cross, and each chord dominates all of the vertices on its zone, if two chords do not share an end, then their zones are disjoint. In particular, we define a chord as *maximal* if its zone is not contained within any other zone.

Now let H be the subgraph induced by all ends of maximal chords and all vertices that lie outside all zones. Note that H is an induced path or cycle. If both ends of a chord dominate all its zone, we call the necessarily unique vertex within the zone a *cap*; if exactly one end of a chord dominates all its zone, we call that end a *pivot*. Note that since vertices in one zone must be able to reach vertices in another zone in three steps, it follows that the pivots form a clique. Also every vertex is within distance two of each pivot and of one neighbor of each cap.

By the diameter condition, the graph H has diameter at most 3. If H is a path, then the two central vertices of H form a TD-set of G . So assume H is a cycle. There are four possibilities.

- (a) If H has seven vertices, then there cannot be any pivots or caps. Thus the total domination number is 4.
- (b) If H has six vertices, then there can be caps but not pivots. So we take any two pairs of vertices on H to form a TD-set of G . (Since opposite edges of such H cannot both

have caps, this case has order at most 9.)

- (c) If H has four or five vertices, we take all but one vertex on H . (To be forced to take four vertices, we therefore need H to have length 5, and it is easily checked that the only possibility is caps on every chord and no pivot, yielding the 5-sun shown in Figure 2.)
- (d) If H has three vertices, we take all of H . But we do not get equality.

This concludes the proof. $\square$

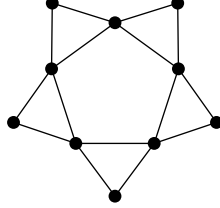


Figure 2: The largest outerplanar graph of diameter 3 with total domination number 4

4.2 Colorings of outerplanar graphs

We start with diameter 2. It follows from Proposition 9 and Theorems 6 and 8, and the fact that outerplanar graphs are 3-colorable, that $\chi_d(G) \leq 5$ and $\chi_{td}(G) \leq 6$ for an outerplanar graph G of diameter 2. But these bounds can be improved, as we show next. Using the fact that $K_{2,3}$ and subdivided K_4 are not outerplanar graphs, it follows readily that the neighbors of a vertex in an outerplanar graph induce a linear forest (disjoint union of paths).

Proposition 11 *If G is an outerplanar graph of diameter at most 2, then $\chi_d(G) \leq 3$ except for the 3-sun (shown on the left in Figure 3), which needs four colors. Furthermore, $\chi_{td}(G) \leq 3$ except for the 5-cycle and the 3-sun, which both need four colors.*

Proof. Consider an embedding of G in the plane such that all vertices lie on the unit circle $\mathcal{C}$ and all edges lie within $\mathcal{C}$. If there is no chord, then the graph is a path or cycle and the result is easily checked (and known). So assume there is a chord st . All other vertices are distance one from (contracted) st ; therefore, they form a linear forest. If G has a universal vertex, then a valid TD-coloring is to give the universal vertex a unique color and give the remaining vertices a bipartite coloring, for a total of three colors. So assume there is no universal vertex.

It follows that there is a vertex s' not adjacent to t and a vertex t' not adjacent to s . Necessarily, vertex s' is adjacent to s and vertex t' is adjacent to t . In particular, vertices s' and t' are in the same linear component, while every vertex z not in that linear component is adjacent to both s and t . By outerplanarity it follows that z is unique, noting that there is no subdivided $K_{2,3}$ with s and t forming the partite set of size 2. Call vertex z a *cap*.

By the above argument, every chord has a cap. By the diameter constraint, any two caps have a common neighbor. Let H be the graph obtained from G by removing the caps: then H is an induced path or cycle. Thus the possibilities are:

- (a) H is a 5-cycle, in which case there cannot be a cap.
- (b) H is a 4-cycle with either one cap (the house graph) or two caps with common neighbor.
- (c) H is a 3-cycle with up to three caps.
- (d) H is a path of length at most 2.

It is easily checked that the only cases that require four colors are the 5-cycle (only in the total version) and the 3-cycle with 3 caps. $\square$

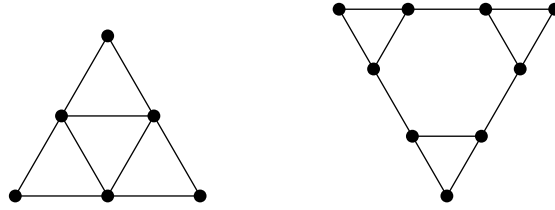


Figure 3: Two outerplanar graphs

Proposition 12 *If G is an outerplanar graph of diameter 3, then $\chi_d(G) \leq 5$. Further, $\chi_{td}(G) \leq 5$ except for the rightmost graph in Figure 3, which needs six colors.*

Proof. Assume first that G has a TD-set J of size at most 3. Such a set necessarily induces a connected subgraph. It follows that $G - J$ is a linear forest; in particular it is bipartite. If we give the vertices of J unique colors and the remainder a bipartite coloring, then we get a valid total coloring using five colors.

So assume that G does not have such a TD-set. It follows from (the proof of) Theorem 10 that there are only a handful of possibilities. We can consider these by hand (or by computer) and find there is only one that needs more than five colors for the total version, namely the rightmost graph in Figure 3. However, this exceptional graph that needs six for a total dominator coloring needs at most five for a dominator coloring. $\square$

5 Bipartite planar graphs of bounded diameter

We start with a discussion of domination (which doesn't seem to have been considered previously). This we then use to provide (possibly not sharp) bounds on the associated chromatic numbers. Theorem 2 follows from the results of this section.

5.1 Domination

Bipartite graphs of diameter 2 are complete bipartite. So the only interesting case is diameter 3. Note that this diameter constraint is equivalent to saying that every two vertices of the same color have a common neighbor.

Theorem 13 *If G is a bipartite planar graph with diameter 3 and bipartition (R, B) , then there exist two vertices of R that dominate all of B , and vice versa. In particular, $\gamma_t(G) \leq 4$.*

Proof. We say a vertex is red if it is in R and blue otherwise. By symmetry it suffices to show that there exists a pair of red vertices that dominates all blue vertices.

Consider an embedding of G in the plane. We say a cycle is *blue-separating* if there is a blue vertex both inside and outside the cycle.

Claim 1 *We may assume there is no blue-separating 4-cycle.*

Proof. Consider a 4-cycle C with blue vertices both inside and outside C . Then the path between a blue inside and a blue outside must go via one of the red vertices on the cycle. So the pair of red vertices on the cycle C dominates all blue vertices and the claim is proved. $\square$

Claim 2 *We may assume there is no blue-separating 6-cycle.*

Proof. Suppose there is a blue-separating 6-cycle C . We may assume it is induced, since a chord would create a blue-separating 4-cycle, a contradiction.

Let C be given by $r_1 b_3 r_2 b_1 r_3 b_2 r_1$. As above, it follows that the triple $\{r_1, r_2, r_3\}$ of red vertices on C dominates the blue vertices. Consider the red pair r_1 and r_2 . Then we are done unless there is a blue vertex x_3 not dominated by r_1 and r_2 . Necessarily, vertex x_3 is adjacent to r_3 . Similarly, we may assume there is a blue vertex x_1 adjacent only to r_1 , and a blue vertex x_2 adjacent only to r_2 . Every pair from the set $\{x_1, x_2, x_3\}$ have a common neighbor. Thus all three of them must be on the same side of C , say inside. Further any blue vertex z outside C is adjacent to all of $\{r_1, r_2, r_3\}$; by planarity, z is unique. See Figure 4.

Consider the blue pair x_1 and b_3 . Since their common neighbor cannot be any of r_1 , r_2 , or r_3 , there is a new red vertex, say y , adjacent to both x_1 and b_3 . The path $r_1 x_1 y b_3$ separates the inside of C . It follows that vertices x_2, b_2, x_3, b_3 are all adjacent to vertex y . But then y is part of a blue-separating 4-cycle, a contradiction. Thus the claim follows. $\square$

So, now consider a pair of red vertices r_1, r_2 that dominates as many of the blue vertices as possible of G . Assume there is a blue vertex b_3 that the pair does not dominate. Let

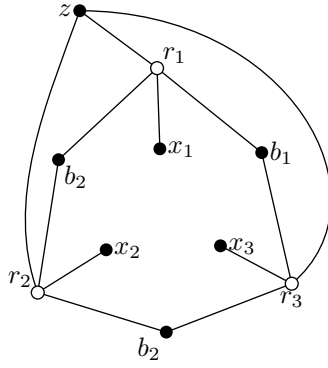


Figure 4: A subgraph containing a blue-separating 6-cycle

x be a common neighbor of vertices r_1 and r_2 . Further, vertices x and b_3 have a common neighbor, say r_3 .

Consider now the pair r_1, r_3 . This pair dominates b_3 ; so by the choice of r_1 and r_2 , there must be a blue vertex b_2 adjacent to r_2 and to neither r_1 nor r_3 . Similarly, there must be a blue vertex b_1 adjacent to r_1 and to neither r_2 nor r_3 .

Now, consider the triple b_1, b_2, b_3 . Assume first that these three vertices have a common neighbor, say y . Then the three paths xr_1b_1y , xr_2b_2y , and xr_3b_3y partition the plane into three regions each bounded by a 6-cycle. Furthermore, for each of these regions there is a blue vertex in the complement of the region. Thus by Claim 2, there is no blue vertex inside that region and hence the graph G contains no blue vertex other than $\{x, b_1, b_2, b_3\}$. It follows that, for example, the pair y and r_1 dominate the blue vertices.

Assume second that the three vertices b_1, b_2, b_3 do not have a common neighbor. Say b_1 and b_2 have common red neighbor y_3 , and define y_1 and y_2 similarly. The six paths xr_1b_1 , xr_2b_2 , xr_3b_3 , $b_1y_3b_2$, $b_2y_1b_3$, and $b_1y_2b_3$ form a subdivided K_4 . As in the previous case, each 6-cycle precludes a blue vertex on one side of the cycle, and these four regions partition the plane. Again it follows that graph G contains no blue vertex other than $\{x, b_1, b_2, b_3\}$. And so for example the pair y_1 and r_1 dominate the blue vertices. $\square$

There are bipartite planar graphs of diameter 3 that need four vertices to dominate. The smallest example is shown in the leftmost graph in Figure 5. The inner trio of degree-2 vertices need at least two vertices to dominate them, as do the outer trio. One can obtain an infinite family by duplicating any degree-2-vertex.

5.2 Colorings

There are bipartite planar graphs of diameter three that need five colors even for ordinary dominator coloring. See the rightmost graph in Figure 5. It follows from the Theorems 8 and 13 that there is an upper bound of 6 on the (total) dominator chromatic number of a

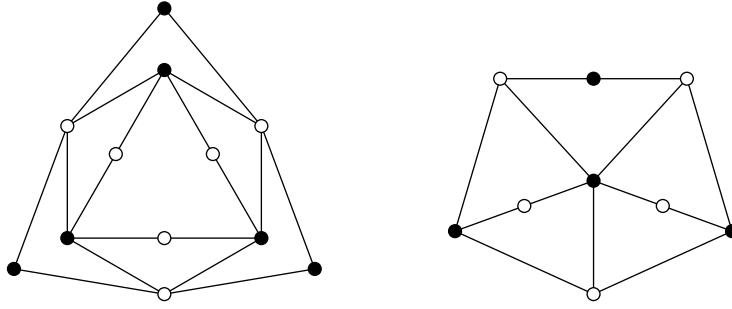


Figure 5: Two bipartite planar graphs of diameter 3

bipartite planar graphs of diameter three. It remains unresolved whether there is a bipartite graph G of diameter 3 with $\chi_d(G) = 6$ or $\chi_{td}(G) = 6$.

6 Planar graphs of diameter two

In this section we consider general planar graphs of diameter 2. By the Four Color Theorem and Theorems 5 and 8, it follows that the (total) dominator chromatic number of such a graph G is at most 7. This upper bound can readily be reduced to 6 by considering a minimum TD-set J and noting that this has size at most 3 by Theorem 5. Furthermore, the set J must induce a connected subgraph and hence $G - J$ is outerplanar and is therefore 3-colorable. It follows from Theorem 8 that $\chi_{td}(G) \leq 6$. But this is not sharp, at least for domination.

Theorem 4 follows immediately from Theorem 3 for $G \neq F$, since a dominator coloring of G is obtained by giving vertices u and v each a unique color and coloring $G - \{u, v\}$ with three colors. It therefore remains to consider the exceptional graph from Theorem 5.

Lemma 14 *If F is the planar graph with diameter 2 and domination number 3, then $\chi_{td}(F) = 4$.*

Proof. The 4-coloring shown in Figure 6 is a dominator coloring of F . It is easily checked that this is best possible. $\square$

We present next an example of a planar graph G with diameter 2 and $\chi_d(G) = 5$. The smallest such graph is shown in Figure 7. To check that four colors are insufficient, let v be the vertex of degree 2 and observe that in any 4-coloring it does not dominate a color class (every color used in its closed neighborhood is also used elsewhere). A suitable 5-coloring is obtained by giving each degree-5 vertex a unique color and the other three vertices the same color.

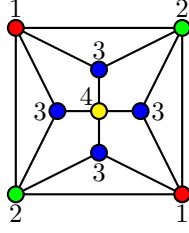


Figure 6: A coloring of the planar graph F of diameter 2 with domination number 3

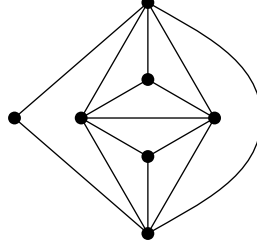


Figure 7: A planar graph G of diameter 2 with $\chi_d(G) = \chi_{td}(G) = 5$

7 Planar graphs of diameter 3

As an immediate consequence of Theorems 7 and 8 and the Four Color Theorem, we have the following result.

Corollary 15 *If G is a planar graph of diameter 3, then the following hold.*

- (a) *We have $\chi_d(G) \leq 13$. Moreover if G has sufficiently large order then $\chi_d(G) \leq 10$.*
- (b) *We have $\chi_d(G) \leq 14$. Moreover if G has sufficiently large order then $\chi_d(G) \leq 11$.*

The 6 in Part 1 of Theorem 7 is sharp. The most we know of is a planar graph G where $\chi_d(G) = \chi_{td}(G) = 9$. (It was inspired by the extremal graphs given in [4].) See Figure 8. To get a suitable coloring, let J be the set of six largest degree vertices (shown in white) and note that J is a dominating set and $G - J$ is 3-colorable. Give each vertex of J a unique color and color the remainder properly. To see that this cannot be improved, even for ordinary domination, note that every degree-2 vertex must have a uniquely colored vertex in its closed neighborhood. This means at least six vertices must get a unique coloring, while the central triangle requires three more colors. Note further that this graph can be grown by duplicating degree-2 vertices.

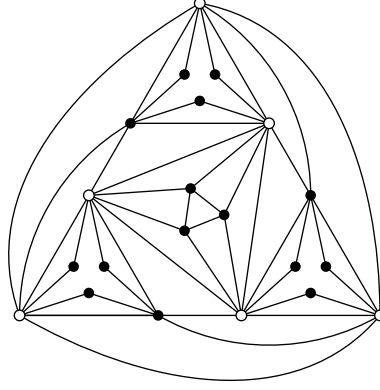


Figure 8: A planar graph G of diameter 3 with $\chi_d(G) = \chi_{td}(G) = 9$

8 Proof of Theorem 3

In this section we prove Theorem 3. Let G be a planar graph of diameter 2 different from the graph F . By Theorem 5 it contains a dominating pair of vertices.

Claim 1 *We may assume that for every dominating pair $\{z_1, z_3\}$, they are nonadjacent and every common neighbor has degree at least 5.*

Proof. If that pair $\{z_1, z_3\}$ is adjacent, then the rest of the graph is outerplanar (if one contracts the pair to a single vertex it dominates) and is therefore 3-colorable, as desired. So assume they are not adjacent and let z_2 be any common neighbor. The subgraph $G - \{z_1, z_2, z_3\}$ is outerplanar and hence 3-colorable. If z_2 has only two neighbors in $G - \{z_1, z_2, z_3\}$, then one can extend the 3-coloring to z_2 , and be done. So we may assume z_2 had at least three neighbors in $G - \{z_1, z_2, z_3\}$ and at least five in total. $\square$

Now fix some dominating pair $\{u, v\}$. Since G has diameter 2, they have at least one common neighbor, say w . Now, consider the embedding of the graph G in the plane. Create graph G' by cutting along the middle of the path uwv to split vertex w into vertices w_1 and w_2 both adjacent to u and v . Note that $H = G' - \{u, v\}$ is outerplanar. So we would be done if there were a 3-coloring of H where w_1 and w_2 receive the same color.

We define a vertex of G as *inner* if it is not adjacent to w . Let I_u and I_v be the inner vertices adjacent to u and to v , respectively. If I_u is empty, then the pair w, v dominates G and is adjacent, a contradiction. So we may assume that both I_u and I_v are nonempty.

8.1 H is 2-connected

Assume first that H is 2-connected. Then, H has an outer hamiltonian cycle. The vertices w_1 and w_2 split the outer cycle into two w_1 - w_2 paths, P_u and P_v , where all of P_u is adjacent

to u and all of P_v is adjacent to v . Note that it also follows that u and v have no other common neighbor.

The diameter requirement means that *every vertex in I_u is adjacent to a neighbor of v and mutatis mutandis*. Because of this property, vertices of I_u and I_v have degree 2 on the graph induced by $V(P_u)$ and $V(P_v)$, respectively. For $i = 1, 2$, define x_i to be the common neighbor of w_i and I_u , and y_i to be the common neighbor of w_i and I_v . The resulting structure of the 2-connected graph H is illustrated in Figure 9.

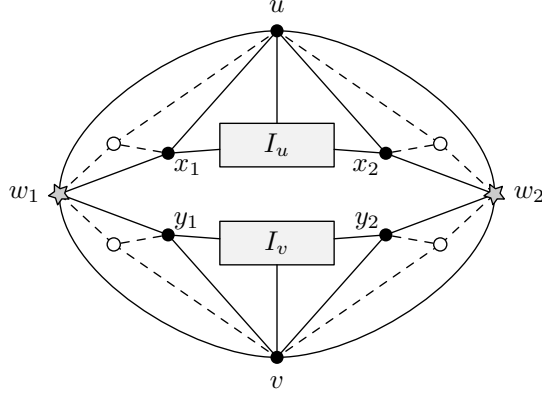


Figure 9: The structure of the graph G' when H is 2-connected

Note that there might or might not be “extra” vertices as shown in white. For example, there might be another common neighbor of w_1 and u , say x'_1 . If so, then it lies inside the triangle w_1, u, x_1 and needs to be adjacent to x_1 to reach I_v in two steps. By planarity it follows that x'_1 would be unique.

Claim 2 *If $|I_u| \geq 3$, then $|I_v| = 1$.*

Proof. Suppose that I_u contains three vertices a, b, c in order (on P_u) and that I_v contains two vertices g, h in order (on P_v).

Both a and b have a neighbor in $N(v)$. There are two cases. The first case is that vertex a is adjacent to vertex y_1 . Then, h and x_1 have a common neighbor; since g has a neighbor in $N(u)$, this common neighbor must be a . But then c does not have a common neighbor with y_1 , a contradiction.

The second case is that vertex a is not adjacent to y_1 . Then the $a-v$ path precludes the edge y_1b . It follows that the common neighbor of y_1 and c must be g . Because a has a neighbor in $N(v)$, it follows that a is adjacent to g . Similarly, b is adjacent to g . But then vertices y_2 and a do not have a common neighbor, a contradiction. $\square$

Claim 3 *It is not possible that $|I_u| = |I_v| = 2$.*

Proof. Assume that the vertices of I_u are a, b in order and the vertices of I_v are g, h in order. Suppose first that extra vertex x'_1 exists. Then its common neighbor with g and h must be x_1 . But then b and y_1 have no common neighbor, a contradiction. Hence, there are no extra vertices and the graph G has order 11.

It cannot be that x_1g and y_1a are both edges. So say x_1 is not adjacent to g . Then g has a neighbor in P_u and so the common neighbor of x_1 and h must be a . It follows that g is adjacent to a . And in fact the diameter-two constraint forces edges y_1a , bh , and hx_2 . Therefore, G' is the graph shown in Figure 10, where edges x_1y_1 and x_2y_2 are both optional. In particular, the pair of vertices w and a dominate G , and have x_1 as a common neighbor of degree at most 4; a contradiction. ($\square$)

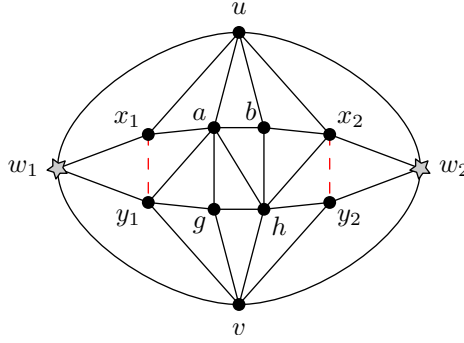


Figure 10: A structure of the graph G' when $|I_u| = |I_v| = 2$

Claim 4 *The result is true if $|I_u| = 1$ and $|I_v| \geq 2$.*

Proof. Assume that I_u has unique vertex a and that the first two vertices of I_v in order are g and h .

Suppose there is an extra vertex adjacent to u . Say there is extra vertex x'_1 adjacent to u and w_1 . Then for it to reach I_v , it must be adjacent to x_1 , and x_1 must be adjacent to all of I_v . For x_2 to reach g , it must be that x_2 is adjacent to h . Thus, I_v has exactly two vertices. Edges x_1y_1 and ah are also necessary. It follows that there are no further extra vertices. Therefore, G' is the graph shown on the left in Figure 11, where the edge x_2y_2 is optional. In particular, vertices x_1 and h dominate G and are adjacent; a contradiction. Hence we may assume there is no extra vertex adjacent to u .

Suppose there is an extra vertex adjacent to v . Say there is extra vertex y'_1 adjacent to v and w_1 . Then, for y'_1 to reach a , it must be that y'_1 is adjacent to y_1 and y_1 is adjacent to a . The common neighbor of h and x_1 must be a . It follows that all of I_v is adjacent to a . We illustrate the resulting graph G' on the right in Figure 11. In particular, the pair of vertices a and v dominate G , and have g as a common neighbor of degree at most 4; a contradiction. Hence we may assume there is no extra vertex at all.

We now rename the vertices of I_v by $g_1g_2 \dots g_j$ (in order on P_v) where $j = |I_v| \geq 2$, and

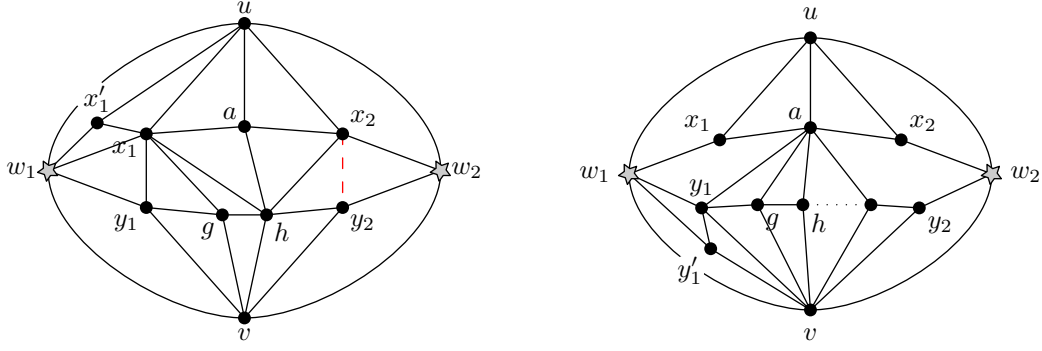


Figure 11: Possible structures of the graph G' when $|I_u| = 1$ and $|I_v| \geq 2$

so $g = g_1$ and $h = g_2$. It cannot be that x_1g_1 and y_1a are both edges. Suppose that x_1 is not adjacent to g_1 . Then, g_1 is adjacent to at least one of a or x_2 . Thus, the common neighbor of x_1 and g_2 must be a . It follows that g_1 is adjacent to a , and indeed that all of I_v is adjacent to a . Thus once again, vertices a and v dominate G , with g_1 as a common neighbor of degree at most 4; a contradiction. Hence, assume x_1g_1 is an edge. Analogously, we may assume x_2g_2 is an edge.

Suppose $j \geq 3$. The vertex a is adjacent to at least one vertex of I_v . Thus in order for x_1 and x_2 to reach y_2 and y_1 , respectively, within two steps, either all of I_v is adjacent to a or $j = 3$ and g_2 is adjacent to all of $\{x_1, a, x_2\}$. If all of I_v is adjacent to a , then as before vertices a and v dominate G , with g_2 as a common neighbor of degree at most 4; a contradiction. Hence, we may assume $j = 3$ and g_2 is adjacent to all of $\{x_1, a, x_2\}$. Thus, in order for a to reach y_1 (resp., y_2) in two steps, we have x_1y_1 and x_2y_2 are edges. The graph G is now determined. Coloring x_1 and x_2 with the color 1, coloring y_1, g_2, y_2 with the color 2, and coloring a, w, g_1, g_3 with the color 3 yields a 3-coloring of $G - \{u, v\}$, as desired. Hence, we may assume that $j = 2$. Further, the graph G has order 10.

By our earlier assumptions, x_1g_1 and x_2g_2 are edges. The vertex a is adjacent to at least one vertex in $I_v = \{g_1, g_2\}$. By symmetry, we may assume ag_1 is an edge. If ag_2 is an edge, then w, u, a is a dominating path P_3 in G where $\{w, a\}$ is a dominating pair and where the vertex u has degree 4 in G , contradicting Claim 1. Hence, ag_2 is not an edge, and so g_2 has degree 4 in G . Thus, $\{g_1, x_2\}$ is a dominating pair in G with common neighbor x_2 of degree 4 in G , a contradiction. This completes the proof of Claim 4. $\square$

So we're left with the case that I_u and I_v have a unique vertex. Note that w dominates everything except the two vertices in $I_u \cup I_v$. Now, if triangle w_1, u, x_1 contains vertex x'_1 , then for the vertex of I_v to reach x'_1 it must be adjacent to x_1 . It follows that w and x_1 are adjacent vertices in G that form a dominating pair, a contradiction. If none of the extra vertices exist, then we have a graph of order 9 and the property is readily confirmed.

8.2 H is not 2-connected

Assume next that H is not 2-connected. Then H has some cut-vertex z .

Claim 5 *Any cut-vertex of H separates w_1 and w_2 .*

Proof. Consider some cut-vertex z . Suppose that w_1 and w_2 are on the same block of H . There there is a w_1 – w_2 path P^* in H not including z . Consider any vertex m of H not on P^* . In order for it to reach both u and v in two steps, it must have a neighbor on P^* , and so is not separated from w_1 and w_2 . That is, every vertex is in the same block as w_1 and w_2 , a contradiction. $\square$

Now, the pair $\{u, v\}$ dominates G and so vertex z has a neighbor in the pair. Moreover, note that even if z and say v are not adjacent, that edge could be added to G while preserving planarity. So it follows that we are done if the pair $\{z, u\}$ or the pair $\{z, v\}$ dominates G , as what remains would be outerplanar.

Claim 6 *In H there exist two w_1 – w_2 paths, say Q_u and Q_v , such that:*

- (i) *every vertex of H is in at least one of the paths;*
- (ii) *every common vertex of the paths, apart from w_1 and w_2 , is a cut-vertex of H ; and*
- (iii) *every vertex on Q_u but not Q_v is adjacent to u , and every vertex on Q_v but not Q_u is adjacent to v .*

Proof. It is well-known that a 2-connected outerplanar graph has a Hamilton cycle. Claim 5 showed that all cut-vertices separate w_1 and w_2 . The result follows by piecing together the Hamilton cycles of the blocks, and noting that the pair $\{u, v\}$ dominates all the vertices. $\square$

We say a cut-vertex z of H is a “switch” if there are non-cut-vertices both before and after z on both Q_u and Q_v .

Claim 7 *The result is true if there is a switch z .*

Proof. Let a be a vertex of H that is not dominated by the pair $\{z, v\}$. Then a lies on Q_u ; say between w_1 and z . On the path Q_v , let m be a non-cut-vertex after z . The common neighbor of m and a in G must be w ; that is, in H vertex m is adjacent to w_2 and vertex a is adjacent to w_1 . Similarly, a vertex of H that is not dominated by the pair $\{z, u\}$ must be adjacent to w_1 or w_2 .

Case 1: *The pair $\{z, w\}$ does not dominate G .*

Say vertex s is adjacent to neither z nor w . Then by the above it must be dominated by both the pair $\{z, v\}$ and the pair $\{z, u\}$. Hence it is a cut-vertex of H . And indeed, vertex s is a switch unless it is joined by a path of length 2 to w , both edges of which are bridges in

H . In that case it is easily checked that H can be 3-colored such that w_1 and w_2 receive the same color. So we may assume that s is a switch. Without loss of generality assume that s is closer to w_1 than z is. Since every vertex between w_1 and s must be adjacent to both of them and to one of $\{u, v\}$, and similarly with the vertices between s and z , and with the vertices between z and w_2 , it follows that H has exactly 10 vertices. See Figure 12. Such an H has a 3-coloring where w_1 and w_2 receive the same color.

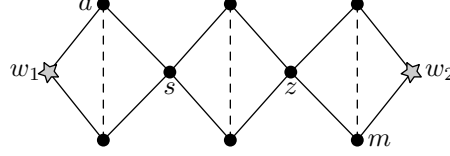


Figure 12: A structure of the graph H

Case 2: *The pair $\{z, w\}$ dominates G .*

It follows in particular that vertices z and w are not adjacent. Now, let b be a vertex that is not dominated by $\{z, u\}$. Necessarily it lies on Q_v .

Case 2.1: Vertex b lies between w_1 and z . Then by the same reasoning as above, the first vertex ℓ on Q_u after z is adjacent to w_2 . Since both ℓ and m are common neighbors of the G -dominating pair $\{z, w\}$, both must have degree at least 5. This forces them to be adjacent, and for there to be a second vertex a' on Q_u and b' on Q_v adjacent to them. But then by the above argument, the situation is symmetric with respect to z . So H is the order-11 graph shown in Figure 13. But in this graph, the 3-coloring of H gives w_1 and w_2 the same color, as desired.

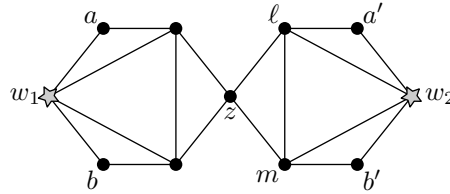


Figure 13: A structure of the graph H

Case 2.2: There is no such b on the segment of Q_v from w_1 to z . So b is on the other segment. But then the vertex before z on Q_u must be adjacent to w . Then by similar reasoning it follows that H is the order-11 graph shown in Figure 14. But in this graph, w_1 and w_2 receive the same color in the 3-coloring of H .

This completes the proof of Claim 7. $\square$

Claim 8 . *The result is true if there is no switch.*

Proof. It follows that the cut-vertex z of H is adjacent to w_1 or w_2 . Suppose there is a second cut-vertex s of H . Then we claim that then H has the desired 3-coloring. For, one

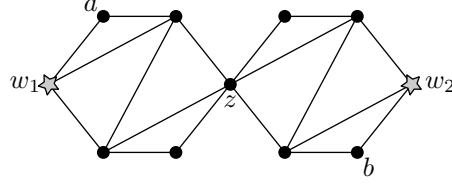


Figure 14: A structure of the graph H

can 3-color the block containing s and z . If c is a/the color that is used at neither s nor z , one can complete the 3-coloring of H giving both w_1 and w_2 color c . So we may assume that z is the unique cut-vertex.

As argued above, neither the pair $\{z, u\}$ nor the pair $\{z, v\}$ dominates G . Therefore we may assume that there is an internal vertex, say q_1 , on the z – w_2 section of the path Q_u and an internal vertex, say q_2 , on the z – w_2 section of the path Q_v such that z is adjacent to neither q_1 nor q_2 . Furthermore, since the pair $\{w, z\}$ does not dominate G , at least one of q_1 and q_2 is not a neighbor of w in G . There are two cases.

Case 1: Both q_1 and q_2 can be chosen nonadjacent to w_2 .

If there is a vertex before z on Q_u (resp., Q_v), then such a vertex would be too far from q_2 (resp., q_1). Hence, there is no such vertex. That is, edge zw_1 is a bridge in H . Suppose that q_1 and q_2 are not adjacent. Then q_1 has a neighbor on Q_v to reach v , while q_2 has a neighbor on Q_u ; from this it follows there are vertices on Q_u and Q_v that are too far apart, a contradiction. (For example, if q_1 has a neighbor before q_2 , then the vertex just before q_1 and the vertex just after q_2 do not have a common neighbor.) Thus we may assume that q_1q_2 is an edge.

Consider the path between the vertex after z on Q_u to the vertex before w on Q_v and the path between the vertex after z on Q_v to the vertex before w on Q_u . Since these paths intersect, it follows readily that at least one of vertices q_1 or q_2 dominates all of H except for $\{w_1, w_2, z\}$: say this is q_1 .

If the pair $\{q_1, v\}$ dominates G , then we are done because they have a common neighbor q_2 of degree 4. It follows that vertex z is adjacent to u but not to v . In order for z to reach all vertices of Q_v , it follows that Q_v has exactly three vertices between z and w_2 . One can also argue that Q_u has exactly three internal vertices. And so the graph H has order 9, as shown in Figure 15. Note that the subgraph H does not have the desired coloring if the dashed edge is present. But it can be checked that the pair $\{w, m\}$ dominates G and the subgraph $G - \{w, m\}$ is 3-colorable.

Case 2: Every choice of q_1 is adjacent to w_2 .

Let p be the vertex before q_1 on Q_u . There are two other possibilities for vertices on Q_u : a vertex n before p and a vertex r after q_1 . The latter must be adjacent to w_2 . See Figure 16. If vertex n is present, then the vertex m on Q_v after q_2 must be adjacent to p to reach n ; similarly, if vertex r is present, then the vertex ℓ on Q_v before q_2 must be adjacent to q_1 to

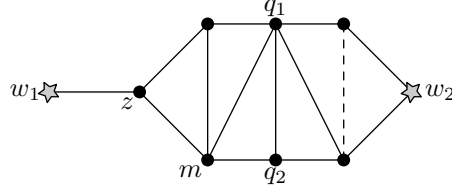


Figure 15: A structure of the graph H

reach r . Since the edges mp and ℓq_1 would have to cross, it follows that at most one of n and r is present.

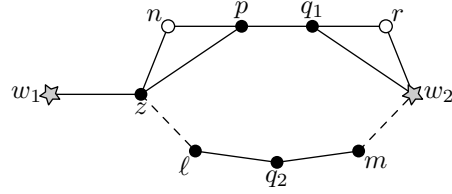


Figure 16: A structure of the graph H

If vertex n is present, then the edge mp exists for all choices of m . Further, the edge q_2p must be present and then the edge ℓp too. It follows that vertex p dominates all of H except w_1, w_2 . For vertex n to reach v it must be that z is adjacent to v . And thus $\{z, p\}$ is a G -dominating pair that are adjacent, a contradiction. Similarly, if vertex r is present, then q_1 dominates all of H except w_1, z and so $\{q_1, w\}$ is a G -dominating pair that are adjacent, a contradiction.

Finally assume neither n nor r is present. There are two cases. Suppose first that there are multiple possibilities for q_2 . Then the farther one must be adjacent to p to reach z . Since all possibilities have to reach u , it follows that all are adjacent to p . Also ℓ is adjacent to p to reach q_1 . See Figure 17. Thus the pair $\{p, w\}$ dominates G and has common neighbor u of degree 4, a contradiction.

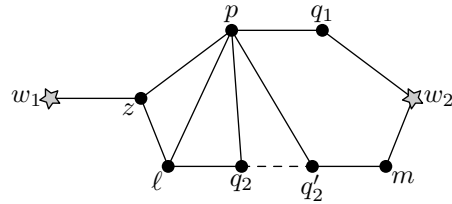


Figure 17: A structure of the graph H

Suppose second that q_2 is unique. Then the graph H has exactly 8 vertices. Vertex q_2 has to reach u . If q_2 is adjacent to p , then because $\{p, w\}$ have common neighbor u of degree 4 in G , then should not dominate G . So p is not adjacent to ℓ . One can check that the missing edge implies that H is 3-colorable with z and w_2 getting different colors, as required. On

the other hand, suppose q_2 is not adjacent to p . Then it is adjacent to q_1 . Because pair $\{q, w\}$ are adjacent in G , they should not dominate G . So q_1 is not adjacent to ℓ . One can check that the missing two edges imply that H is 3-colorable with z and w_2 getting different colors. This completes the proof of Claim 8 and the proof of the Theorem. ($\square$)

This completes the proof of Theorem 3. $\square$

We suspect that a similar, though more lengthy, approach could be used to show that in fact the total dominator chromatic number of a diameter-2 planar graph is also at most 5.

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